

Example 2 HW

The voltage and current in a circuit are given by $\bar{V} = 150 \angle 30^\circ \text{ V}$ and $\bar{I} = 2 \angle -15^\circ \text{ A}$. If the circuit works on a 50 Hz supply, determine impedance, resistance, reactance, power factor and power loss considering the circuit as a simple series circuit.

Solution

$$\bar{V} = 150 \angle 30^\circ \text{ V}$$

$$\bar{I} = 2 \angle -15^\circ \text{ A}$$

$$f = 50 \text{ Hz}$$

(i) Impedance

$$\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{150 \angle 30^\circ}{2 \angle -15^\circ} = 75 \angle 45^\circ \Omega = 53.03 + j53.03 \Omega$$

$$Z = 75 \Omega$$

(ii) Resistance

$$R = 53.03 \Omega$$

(iii) Reactance

$$X = 53.03 \Omega$$

(iv) Power factor

$$\phi = 45^\circ$$

$$\text{pf} = \cos \phi = \cos (45^\circ) = 0.707 \text{ (lagging)}$$

(v) Power loss

$$P = VI \cos \phi = 150 \times 2 \times 0.707 = 212.1 \text{ W}$$

Example 3 HW

An rms voltage of $100 \angle 0^\circ \text{ V}$ is applied to a series combination of Z_1 and Z_2 when $Z_1 = 20 \angle 30^\circ \Omega$. The effective voltage drop across Z_1 is known to be $40 \angle -30^\circ \text{ V}$. Find the reactive component of Z_2 .

Solution

$$\bar{V} = 100 \angle 0^\circ \text{ V}$$

$$\bar{Z}_1 = 20 \angle 30^\circ \Omega$$

$$\bar{V}_1 = 40 \angle -30^\circ \text{ V}$$

$$\bar{I} = \frac{\bar{V}_1}{\bar{Z}_1} = \frac{40 \angle -30^\circ}{20 \angle 30^\circ} = 2 \angle -60^\circ \text{ A}$$

$$\bar{Z} = \frac{\bar{V}}{\bar{I}} = \frac{100 \angle 0^\circ}{2 \angle -60^\circ} = 50 \angle 60^\circ = 25 + j43.3 \Omega$$

$$\bar{Z}_1 = 20 \angle 30^\circ = 17.32 + j10 \Omega$$

$$\bar{Z} = \bar{Z}_1 + \bar{Z}_2$$

$$\bar{Z}_2 = \bar{Z} - \bar{Z}_1 = 25 + j43.3 - 17.32 - j10 = 7.68 + j33.3 \Omega$$

Reactive component of $\bar{Z}_2 = 33.3 \Omega$

H/W
Example 4

A voltage $v(t) = 177 \sin(314t + 10^\circ)$ is applied to a circuit. It causes a steady-state current to flow, which is described by $i(t) = 14.14 \sin(314t - 20^\circ)$. Determine the power factor and average power delivered to the circuit.

Solution

$$v(t) = 177 \sin(314t + 10^\circ)$$

$$i(t) = 14.14 \sin(314t - 20^\circ)$$

(i) Power factor

Current $i(t)$ lags behind voltage $v(t)$ by 30° .

$$\phi = 30^\circ$$

$$\text{pf} = \cos \phi = \cos(30^\circ) = 0.866 \text{ (lagging)}$$

(ii) Average power

$$P = VI \cos \phi = \frac{177}{\sqrt{2}} \times \frac{14.14}{\sqrt{2}} \times 0.866 = 1083.7 \text{ W}$$

class
Example 5

When a sinusoidal voltage of 120 V (rms) is applied to a series R-L circuit, it is found that there occurs a power dissipation of 1200 W and a current flow given by $i(t) = 28.3 \sin(314t - \phi)$. Find the circuit resistance and inductance.

Solution

$$V = 120 \text{ V}$$

$$P = 1200 \text{ W}$$

$$i(t) = 28.3 \sin(314t - \phi)$$

(i) Resistance

$$I = \frac{28.3}{\sqrt{2}} = 20.01 \text{ A}$$

$$P = VI \cos \phi$$

$$1200 = 120 \times 20.01 \times \cos \phi$$

$$\cos \phi = 0.499$$

$$\phi = 60.02^\circ$$

$$Z = \frac{V}{I} = \frac{120}{20.01} = 6 \Omega$$

$$\bar{Z} = Z \angle \phi = 6 \angle 60.02^\circ = 3 + j5.2 \Omega$$

$$R = 3 \Omega$$

(ii) Inductance

$$X_L = 5.2 \Omega$$

$$X_L = \omega L$$

$$5.2 = 314 \times L$$

$$L = 0.0165 \text{ H}$$

HW Example 6

In a series circuit containing resistance and inductance, the current and voltage are expressed as $i(t) = 5 \sin\left(314t + \frac{2\pi}{3}\right)$ and $v(t) = 20 \sin\left(314t + \frac{5\pi}{6}\right)$. (i) What is the impedance of the circuit? (ii) What are the values of resistance, inductance and power factor? (iii) What is the average power drawn by the circuit?

Solution

$$i(t) = 5 \sin\left(314t + \frac{2\pi}{3}\right)$$

$$v(t) = 20 \sin\left(314t + \frac{5\pi}{6}\right)$$

(i) Impedance

$$Z = \frac{V}{I} = \frac{V_m}{I_m} = \frac{20}{5} = 4 \Omega$$

(ii) Power factor, resistance and inductance

Current $i(t)$ lags behind voltage $v(t)$ by an angle $\phi = 150^\circ - 120^\circ = 30^\circ$

$$\text{pf} = \cos \phi = \cos(30^\circ) = 0.866 \text{ (lagging)}$$

$$\bar{Z} = 4 \angle 30^\circ = 3.464 + j2 \Omega$$

$$R = 3.464 \Omega$$

$$X_L = 2 \Omega$$

$$X_L = \omega L$$

$$2 = 314 \times L$$

$$L = 6.37 \text{ mH}$$

(iii) Average power

$$P = VI \cos \phi = \frac{20}{\sqrt{2}} \times \frac{5}{\sqrt{2}} \times 0.866 = 43.3 \text{ W}$$

HW Example 7

A series circuit consists of a non-inductive resistance of 6Ω and an inductive reactance of 10Ω . When connected to a single-phase ac supply, it draws a current $i(t) = 27.89 \sin(628t - 45^\circ)$. Calculate (i) the voltage applied to the series circuit in the form $V_m \sin(\omega t \pm \phi)$, (ii) inductance, and (iii) power drawn by the circuit.

Solution

$$R = 6 \Omega$$

$$X_L = 10 \Omega$$

$$i(t) = 27.89 \sin(628t - 45^\circ)$$

(i) Voltage applied to the series circuit

$$\bar{Z} = R + jX_L = 6 + j10 = 11.66 \angle 59.04^\circ \Omega$$

$$\bar{I} = \frac{27.89}{\sqrt{2}} \angle -45^\circ = 19.72 \angle -45^\circ \text{ A}$$

$$\bar{V} = \bar{Z} \bar{I} = (11.66 \angle 59.04^\circ) (19.72 \angle -45^\circ) = 229.95 \angle 14.04^\circ \text{ V}$$

$$v = 229.95 \sqrt{2} \sin(\omega t + 14.04^\circ) = 325.2 \sin(\omega t + 14.04^\circ)$$

(ii) Inductance

$$X_L = \omega L$$

$$10 = 628 \times L$$

$$L = 15.9 \text{ mH}$$

(iii) Power drawn by the circuit

$$P = VI \cos \phi = 229.95 \times 19.72 \times \cos(59.04^\circ) = 2332.78 \text{ W}$$

Example 8

A coil having a resistance of 10Ω and an inductance of 40 mH is connected to a 200 V , 50 Hz supply. Calculate the impedance of the coil, current, power factor and power consumed. [May 2015]

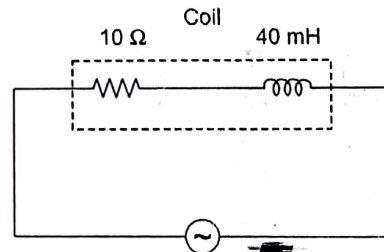
Solution $r = 10 \Omega$

$$L = 40 \text{ mH}$$

$$V = 200 \text{ V}$$

$$f = 50 \text{ Hz}$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 40 \times 10^{-3} = 12.57 \Omega$$



200 V, 50 Hz

Fig. 4.18

(i) Impedance of coil

$$\bar{Z}_{\text{coil}} = r + jX_L = 10 + j12.57 = 16.06 \angle 51.5^\circ \Omega$$

(ii) Current

$$I = \frac{V}{Z_{\text{coil}}} = \frac{200}{16.06} = 12.45 \text{ A}$$

(iii) Power factor

$$\text{pf} = \cos \phi = \cos(51.5^\circ) = 0.62 \text{ (lagging)}$$

(iv) Power consumed

$$P = VI \cos \phi = 200 \times 12.45 \times 0.62 = 1554 \text{ W} = 1.554 \text{ kW}$$

Example 9

When an inductive coil is connected to a dc supply at 240 V, the current in it is 16 A. When the same coil is connected to an ac supply at 240 V, 50 Hz, the current is 12.27 A. Calculate (i) resistance, (ii) impedance, (iii) reactance, and (iv) inductance of the coil.

Solution For dc: $V = 240 \text{ V}$, $I = 16 \text{ A}$

For ac: $V = 240 \text{ V}$, $I = 12.27 \text{ A}$

(i) Resistance

When an inductive coil is connected to a dc supply,

$$f = 0$$

$$X_L = 2\pi fL = 0$$

The coil behaves like a pure resistor.

$$R = \frac{V}{I} = \frac{240}{16} = 15 \Omega$$

(ii) Impedance

When the coil is connected to an ac supply,

$$Z = \frac{V}{I} = \frac{240}{12.27} = 19.56 \Omega$$

(iii) Reactance

$$X_L = \sqrt{Z^2 - R^2} = \sqrt{(19.56)^2 - (15)^2} = 12.55 \Omega$$

(iv) Inductance

$$X_L = 2\pi fL$$

$$12.55 = 2\pi \times 50 \times L$$

$$L = 0.04 \text{ H}$$

Example 10

An inductive coil draws 10 A current and consumes 1 kW power from a 200 V, 50 Hz ac supply. Determine (i) impedance in Cartesian and polar forms, (ii) power factor, and (iii) reactive and apparent power.

Solution

$$I = 10 \text{ A}$$

$$P = 1 \text{ kW}$$

$$V = 200 \text{ V}$$

$$f = 50 \text{ Hz}$$

(i) Impedance in Cartesian and polar forms

$$Z = \frac{V}{I} = \frac{200}{10} = 20 \Omega$$

$$P = VI \cos \phi$$

$$1000 = 200 \times 10 \times \cos \phi$$

$$\cos \phi = 0.5$$

$$\phi = 60^\circ$$

Expressing $\bar{Z}$ in polar form,

$$\bar{Z} = Z \angle \phi^\circ = 20 \angle 60^\circ \Omega$$

Expressing $\bar{Z}$ in Cartesian form,

$$\bar{Z} = 10 + j17.32 \Omega$$

(ii) Power factor

$$\text{pf} = \cos \phi = \cos (60^\circ) = 0.5 \text{ (lagging)}$$

(iii) Reactive and apparent power

$$Q = VI \sin \phi = 200 \times 10 \times \sin (60^\circ) = 1.73 \text{ kVAR}$$

$$S = VI = 200 \times 10 = 2 \text{ kVA}$$

Example 11

A coil connected across a 250 V, 50 Hz supply takes a current of 10 A at 0.8 lagging power factor. What will be the power taken by the choke coil when connected across a 200 V, 25 Hz supply? Also calculate resistance and inductance of the coil.

Solution

$$V_1 = 250 \text{ V}$$

$$f_1 = 50 \text{ Hz}$$

$$I_1 = 10 \text{ A}$$

$$\text{pf} = 0.8 \text{ (lagging)}$$

$$V_2 = 200 \text{ V}$$

$$f_2 = 25 \text{ Hz}$$

(i) Resistance and inductance of the coil

$$Z_1 = \frac{V_1}{I_1} = \frac{250}{10} = 25 \Omega$$

$$\phi_1 = \cos^{-1} (0.8) = 36.87^\circ$$

$$\bar{Z}_1 = Z_1 \angle \phi_1 = 25 \angle 36.87^\circ = 20 + j15 \Omega$$

$$R = 20 \Omega$$

$$X_{L_1} = 15 \Omega$$

$$X_{L_1} = 2\pi f_1 L$$

$$15 = 2\pi \times 50 \times L$$

$$L = 0.0477 \text{ H}$$

(ii) Power taken by the choke coil when connected across a 200 V, 50 Hz supply.

$$X_{L_2} = 2\pi f_2 L = 2\pi \times 25 \times 0.0477 = 7.49 \Omega$$

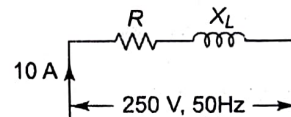


Fig. 4.19